

CSE 107 - Homework 4 Solutions

Problem 1: Hough Transform

Part (a): Equation for the Sinusoid at Pixel Location (o,o)

The Hough Transform converts points from image space to parameter space using the normal form of a line:

$$\rho = x \cos \theta + y \sin \theta$$

where:

- ρ is the perpendicular distance from the origin to the line
- θ is the angle of the perpendicular from the origin

For an edge point at pixel location (o,o):

Substituting $x = 0$ and $y = 0$ into the equation:

$$\rho = 0 \cdot \cos \theta + 0 \cdot \sin \theta$$

$$\boxed{\rho = 0}$$

This means the sinusoid corresponding to the point (o,o) is simply the horizontal line $\rho = 0$ in parameter space. This line extends across all values of θ from 0 to π (or $-\pi/2$ to $\pi/2$ depending on the convention).

Part (b): Other Points Corresponding to This Sinusoid

Answer: No, there are no other points (at locations other than (o,o)) that correspond to this particular sinusoid.

Explanation:

The sinusoid for any point (x, y) in image space is given by:

$$\rho = x \cos \theta + y \sin \theta$$

For this equation to produce $\rho = 0$ for **all values of θ** (which is what the sinusoid from $(0,0)$ produces), we would need:

$$x \cos \theta + y \sin \theta = 0 \quad \forall \theta$$

This condition can only be satisfied when both $x = 0$ AND $y = 0$.

For any other point $(x, y) \neq (0, 0)$, the equation produces a true sinusoidal curve that varies with θ , oscillating between positive and negative values of ρ . While these other sinusoids will *intersect* the line $\rho = 0$ at specific values of θ , they do not *coincide* with it across all θ .

Key Insight: The point $(0,0)$ is unique because all lines passing through the origin have $\rho = 0$, so its "sinusoid" degenerates into a horizontal line in parameter space.

Problem 2: Region Splitting and Merging (Quadtree Segmentation)

Given:

- Image size: $N = 8$ (8×8 pixels)
- Predicate: $Q(R_i) = \text{TRUE}$ if all pixels in R_i have the same intensity
- Image has two intensity values (white and gray)

Binary Representation of the Image:

Row\Col	0	1	2	3	4	5	6	7
0	W	W	W	W	W	W	W	W
1	W	W	W	W	W	W	W	W
2	W	W	G	G	G	G	W	W
3	W	W	G	W	W	G	W	W
4	W	W	G	W	W	G	W	W
5	W	W	G	G	G	G	W	W
6	W	W	W	W	W	W	W	W
7	W	W	W	W	W	W	W	W

W = White (intensity 1)

G = Gray (intensity 0)

Quadtree Splitting Process:

Level 0 (Root): R - Entire 8×8 image

- Contains both white and gray pixels
- $Q(R) = \text{FALSE} \rightarrow \text{SPLIT}$

Level 1: Split into four 4×4 quadrants

Quadrant	Position	Contents	$Q(R_i)$	Action
R_1	Top-left (rows 0-3, cols 0-3)	Mixed (W and G)	FALSE	SPLIT
R_2	Top-right (rows 0-3, cols 4-7)	Mixed (W and G)	FALSE	SPLIT
R_3	Bottom-left (rows 4-7, cols 0-3)	Mixed (W and G)	FALSE	SPLIT
R_4	Bottom-right (rows 4-7, cols 4-7)	Mixed (W and G)	FALSE	SPLIT

Level 2: Split each 4×4 quadrant into 2×2 sub-quadrants

R_1 Sub-quadrants:

Region	Position	Contents	$Q(R_i)$	Action
R_{11}	rows 0-1, cols 0-1	All White	TRUE	STOP
R_{12}	rows 0-1, cols 2-3	All White	TRUE	STOP
R_{13}	rows 2-3, cols 0-1	All White	TRUE	STOP
R_{14}	rows 2-3, cols 2-3	Mixed (G and W)	FALSE	SPLIT

R_2 Sub-quadrants:

Region	Position	Contents	$Q(R_i)$	Action
R_{21}	rows 0-1, cols 4-5	All White	TRUE	STOP
R_{22}	rows 0-1, cols 6-7	All White	TRUE	STOP
R_{23}	rows 2-3, cols 4-5	Mixed (G and W)	FALSE	SPLIT
R_{24}	rows 2-3, cols 6-7	All White	TRUE	STOP

R_3 Sub-quadrants:

Region	Position	Contents	$Q(R_i)$	Action
R_{31}	rows 4-5, cols 0-1	All White	TRUE	STOP
R_{32}	rows 4-5, cols 2-3	Mixed (G and W)	FALSE	SPLIT
R_{33}	rows 6-7, cols 0-1	All White	TRUE	STOP
R_{34}	rows 6-7, cols 2-3	All White	TRUE	STOP

R_4 Sub-quadrants:

Region	Position	Contents	$Q(R_i)$	Action
R_{41}	rows 4-5, cols 4-5	Mixed (G and W)	FALSE	SPLIT

R_{42}	rows 4-5, cols 6-7	All White	TRUE	STOP
R_{43}	rows 6-7, cols 4-5	All White	TRUE	STOP
R_{44}	rows 6-7, cols 6-7	All White	TRUE	STOP

Level 3: Split remaining 2×2 regions into 1×1 pixels

R_{14} (rows 2-3, cols 2-3): R_{141} : (2,2) = G, R_{142} : (2,3) = G, R_{143} : (3,2) = G, R_{144} : (3,3) = W

R_{23} (rows 2-3, cols 4-5): R_{231} : (2,4) = G, R_{232} : (2,5) = G, R_{233} : (3,4) = W, R_{234} : (3,5) = G

R_{32} (rows 4-5, cols 2-3): R_{321} : (4,2) = G, R_{322} : (4,3) = W, R_{323} : (5,2) = G, R_{324} : (5,3) = G

R_{41} (rows 4-5, cols 4-5): R_{411} : (4,4) = W, R_{412} : (4,5) = G, R_{413} : (5,4) = G, R_{414} : (5,5) = G

Part (a): Segmented Image with Labeled Quadrants

Col:	0	1	2	3	4	5	6	7
Row 0	R_{11}		R_{12}		R_{21}		R_{22}	
Row 1	(W)		(W)		(W)		(W)	
Row 2	R_{13}	R_{141}	R_{142}		R_{231}	R_{232}	R_{24}	
Row 3		R_{143}	R_{144}		R_{233}	R_{234}		
		G	G		G	G		
		G	W		W	G		
Row 4	R_{31}	R_{321}	R_{322}		R_{411}	R_{412}	R_{42}	
Row 5		R_{323}	R_{324}		R_{413}	R_{414}		
		G	G		G	G		
Row 6	R_{33}		R_{34}		R_{43}		R_{44}	
Row 7	(W)		(W)		(W)		(W)	

Part (b): Quadtree Structure

R (8×8) – Mixed | R_1 (4×4) – Mixed | R_{11} (2×2) – WHITE ✓ | R_{12} (2×2) – WHITE ✓ | R_{13} (2×2) – WHITE ✓ | R_{14} (2×2) – Mixed | R_{141} (1×1) – GRAY ✓ | R_{142} (1×1) – GRAY ✓ | R_{143} (1×1) – GRAY ✓ | R_{144} (1×1) – WHITE ✓ | R_2 (4×4) – Mixed | R_{21} (2×2) – WHITE ✓ | R_{22} (2×2) – WHITE ✓ | R_{23} (2×2) – Mixed | R_{231} (1×1) – GRAY ✓ | R_{232} (1×1) –

GRAY ✓ | | | — R_{2 3 3} (1×1) — WHITE ✓ | | | — R_{2 3 4} (1×1) — GRAY ✓ | | — R_{2 4} (2×2) — WHITE ✓ | — R₃ (4×4) — Mixed | | — R_{3 1} (2×2) — WHITE ✓ | | — R_{3 2} (2×2) — Mixed | | | — R_{3 2 1} (1×1) — GRAY ✓ | | | — R_{3 2 2} (1×1) — WHITE ✓ | | | — R_{3 2 3} (1×1) — GRAY ✓ | | | — R_{3 2 4} (1×1) — GRAY ✓ | | — R_{3 3} (2×2) — WHITE ✓ | | — R_{3 4} (2×2) — WHITE ✓ | — R₄ (4×4) — Mixed | — R_{4 1} (2×2) — Mixed | | — R_{4 1 1} (1×1) — WHITE ✓ | | — R_{4 1 2} (1×1) — GRAY ✓ | | — R_{4 1 3} (1×1) — GRAY ✓ | | — R_{4 1 4} (1×1) — GRAY ✓ | — R_{4 2} (2×2) — WHITE ✓ | — R_{4 3} (2×2) — WHITE ✓ | — R_{4 4} (2×2) — WHITE ✓

Summary:

- Total leaf nodes: 28
- White regions: 16 (12 at 2×2 level + 4 at 1×1 level)
- Gray regions: 12 (all at 1×1 level)

Problem 3: Watershed Segmentation

Given Cross-Section Data:

From the figure, the intensity values at each position are:

Position (x)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Intensity	0	0	3	7	5	4	6	5	0	1	3	1	2	2	0

Identifying Catchment Basins (Local Minima):

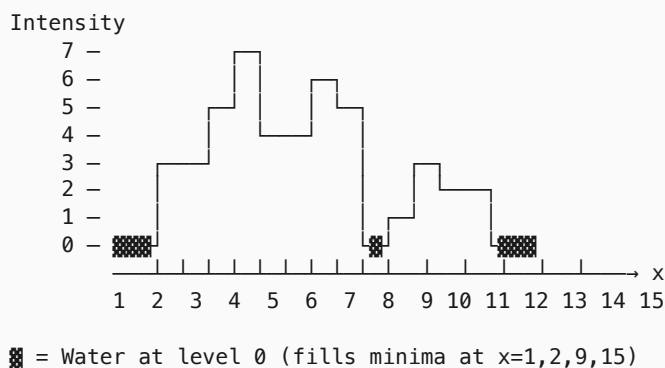
Local minima locations:

1. **Basin 1:** $x = 1-2$ (intensity 0) - Left edge minimum
2. **Basin 2:** $x = 9$ (intensity 0) - Central minimum
3. **Basin 3:** $x = 15$ (intensity 0) - Right edge minimum

Step-by-Step Watershed Algorithm:

The algorithm "floods" from the bottom up, filling from local minima. Dams are built where water from different basins would meet.

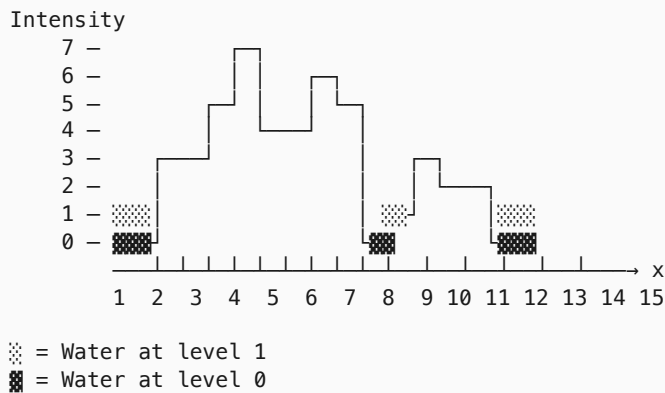
Step 0: Initial State (Water Level = 0)



Three basins begin filling:

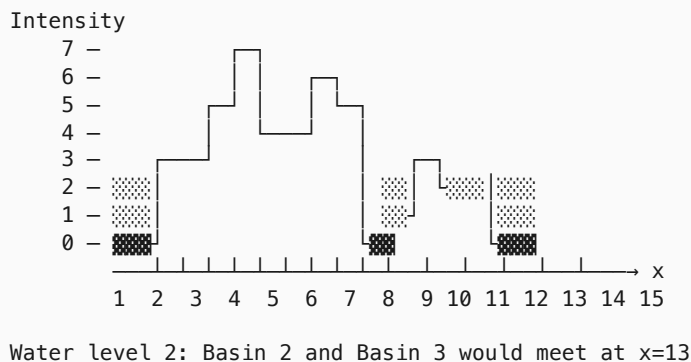
- Basin 1: $x = 1, 2$
- Basin 2: $x = 9$
- Basin 3: $x = 15$

Step 1: Water Level = 1



- Basin 1 expands slightly (still $x = 1, 2$)
- Basin 2 expands to $x = 9, 10, 12$ (since intensity at $x=10$ and $x=12$ is 1)
- Basin 3 expands to $x = 14, 15$

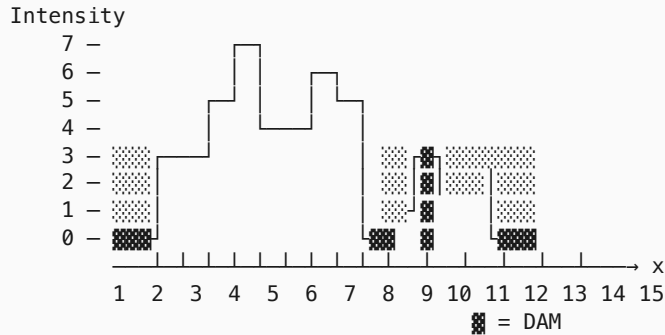
Step 2: Water Level = 2



DAM 1 built between positions 12 and 13 to separate Basin 2 from Basin 3.

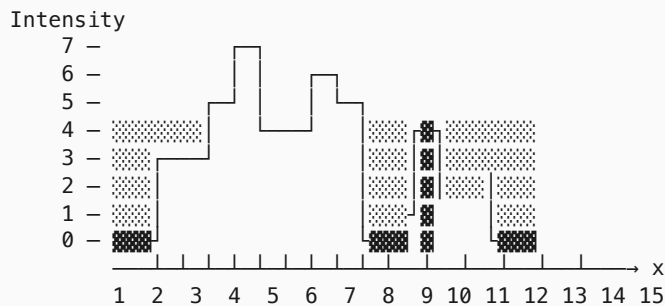
- Basin 2: $x = 9, 10, 11, 12$
- Basin 3: $x = 13, 14, 15$

Step 3: Water Level = 3



- Basin 1: $x = 1, 2, 3$
- Basin 2: $x = 9, 10, 11, 12$ (blocked by dam at right)
- Basin 3: $x = 13, 14, 15$ (blocked by dam at left)

Step 4: Water Level = 4

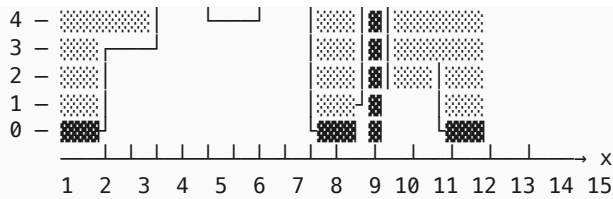


- Basin 1: $x = 1, 2, 3$ (extends toward peak at $x=4$)
- Basin 2: $x = 6, 8, 9, 10, 11, 12$ (extends from $x=6$ and $x=8$)

Basin 1 and Basin 2 approaching each other...

Step 5: Water Level = 5

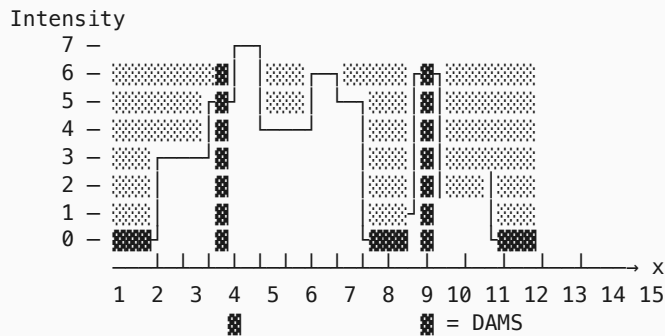




Water from Basin 1 and Basin 2 would meet!

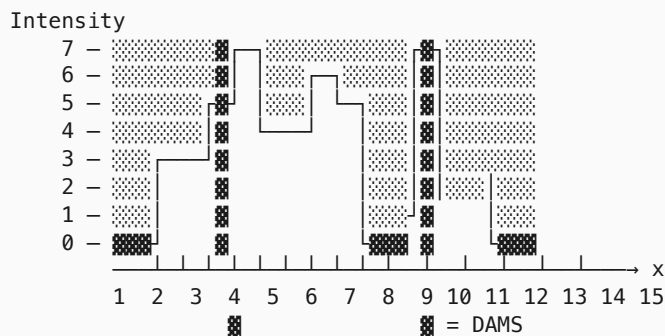
DAM 2 built between positions 4 and 5 (at the ridge between the two basins)

Step 6: Water Level = 6



Basin 2 fills around the internal ridge at $x=7$. This is within the same basin, so no new dam needed.

Step 7: Water Level = 7 (Final)



Final Segmentation Result:

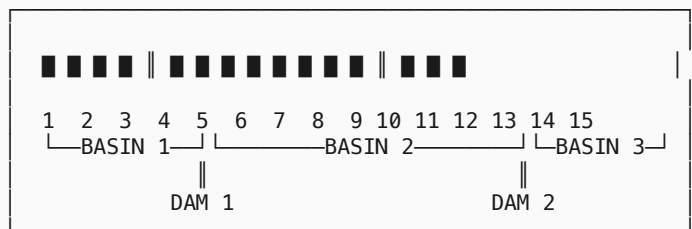
Three watershed regions (catchment basins) identified:

Segment	Positions	Description
Basin 1	$x = 1, 2, 3, 4$	Left region
Basin 2	$x = 5, 6, 7, 8, 9, 10, 11, 12$	Central region
Basin 3	$x = 13, 14, 15$	Right region

Two dams (watershed lines) constructed:

Dam	Location	Separates
Dam 1	Between $x = 4$ and $x = 5$	Basin 1 and Basin 2
Dam 2	Between $x = 12$ and $x = 13$	Basin 2 and Basin 3

Summary Diagram:



The watershed segmentation successfully divides the 1D intensity profile into three distinct regions based on the local minima and the construction of dams at the points where water from different catchment basins would merge.

End of Homework 4 Solutions